SECTION 9 – MAGNET RAMPING PROCEDURE

Description

The 1.0T Cryostable magnet, with serial number prefix "L", Ramping current is approximately 740 amps. This procedure utilizes Current Control for magnet ramping. If Ramp and Shim Cables were not ordered with the magnet, Ramp Cable Kit P/N 2135435 will be needed to ramp the magnet, and Shim Cable Kit P/N 2135558 will be needed to shim the magnet.

WARNING!

THE MRU CIRCUIT FOR THE ALTERNATE CONFIGURATION MAGNETS (MAGNET MODEL NUMBER 2141397–2) IS THROUGH THE SHIM LEAD. MAKE SURE THE MRU IS INSTALLED AND TESTED IN CONFORMANCE WITH THE VENDOR MANUAL BEFORE RAMPING THE MAGNET.

Note

All references in this procedure are to *Direction 15537,* GE 1.0T MAGNET AND CRYOGEN SUBSYSTEM.

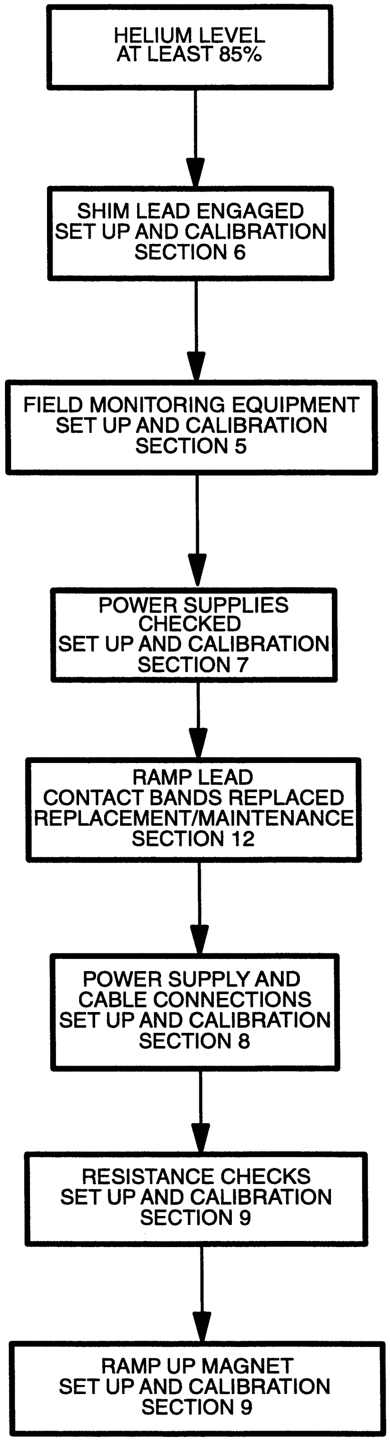
WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

1. **MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5–3 OF THE INTRODUCTION (CRYOGEN SAFETY).**
2. **NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.**
3. **POST WARNING SIGNS OUTSIDE THE 5 GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE INTRODUCTION, SECTION 5.**

SECTION 9 – MAGNET RAMPING PROCEDURE (CONTINUED)

4. POST "RAMPED MAGNET" WARNING SIGN AT THE MAGNET ROOM ENTRANCE PRIOR TO RAMPING THE MAGNET. WARNING TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.
5. REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.
6. MAKE SURE THAT THE MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET UP AND CALIBRATION, SECTION 1–5.
7. MAKE SURE THAT THE MAGNET IS AT LEAST 85% FULL OF LIQUID HELIUM TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.



RAMP FLOWCHART
ILLUSTRATION 9–1

9–1 PREPARATION

1. Set up the field monitoring equipment, Teslameter and Teslameter Probe, in conformance with SET UP AND CALIBRATION, Section 5.
2. Perform Magnet Electrical Checks in conformance with FUNCTIONAL CHECKS, Section 3.
3. Make sure the magnet is at least 85% full of helium before Ramping the magnet.

WARNING!

THE 1.0T MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND/OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS ROUTINELY CALIBRATED AT AN APPROVED FACILITY.

4. Make sure that the Main Power Supply is installed, checked and adjusted in conformance with the Vendor Manual supplied with the unit.
5. Make sure the Shim Lead Assembly is "Engaged" in conformance with SET UP AND CALIBRATION, Section 6.
6. Make sure that AC Input Power Cable to the Power Supply is disconnected.
7. Connect the Power Supply to the magnet by making all cable connections in conformance with SET UP AND CALIBRATION, Section 8. ("Electrical Connections For Ramping And Shimming")
8. Record the Main Coil connection polarity in DATA SHEETS, Table 6–1.
9. Set all power supply heater switches to the OFF position.
10. Set CURRENT ADJUST and VOLTAGE controls to 0 (full CCW).
11. Connect the AC Input Power Cable to the Main Power Supply.
12. Make sure He Vent Valve (V2) is closed.

9–2 RESISTANCE CHECKS

1. Make sure CURRENT ADJUST and VOLTAGE ADJUST controls on the Main Power Supply are off (full CCW).

Note

All "Main Coil Driving Voltages" provided in this procedure will be equal in magnitude but opposite in polarity for "Reverse Ramped" magnets.

WARNING!

MAKE SURE MAIN HEATER SWITCH IS OFF DURING THE RESISTANCE CHECKS.

2. Set the MAIN POWER and POWER ON switches to ON (Switches located on both the front and back of the Main Power Supply).

9–2 RESISTANCE CHECKS (continued)

3. Set HEATER 2 SHIM AXIAL switch to 1 (on) and observe current rise in ammeter (800–820 mA) to verify circuit continuity. Make sure Main Heater Switch is off.
4. Connect a Digital Voltmeter (DVM) to the end of the Voltage Sense Leads.
5. Set CURRENT ADJUST COARSE control on power supply to maximum (full CW).
6. Observe the Main Power Supply Ammeter and slowly turn the VOLTAGE control (CW) to set 750A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
7. Record the voltage reading on the (DVM) in the DATA SHEET tab, Table 6–1.

WARNING!

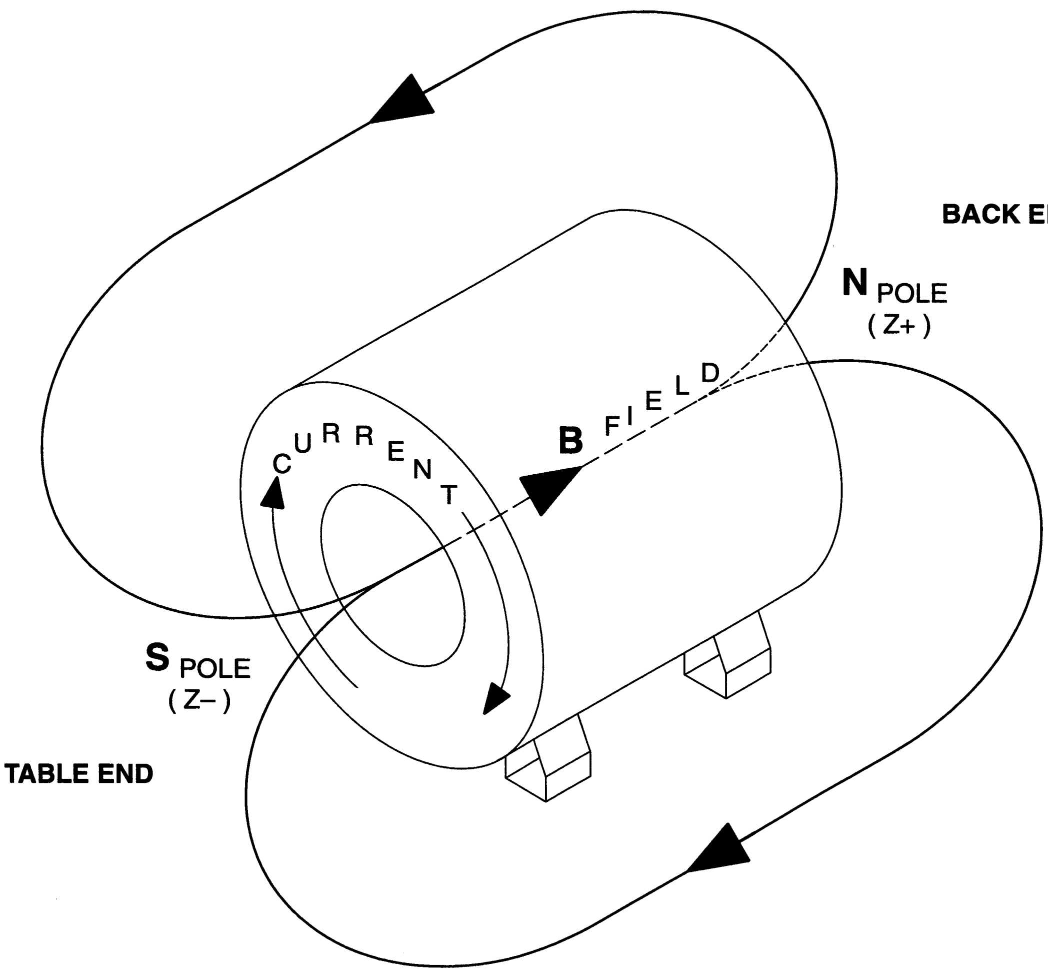
A VOLTAGE READING GREATER THAN 150 MILLIVOLTS AT 750 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET INCREASING BOILOFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.

8. Perform one or more of the bulleted steps below, as necessary, if the DVM voltage is greater than 150mV.
   - Wait approximately 1 minute with the current running, readings may drop as the Power Lead Extensions cool.
   - Tighten the nuts on top of the Hold Down Tool.
   - If the reading still exceeds 150 mV: turn the VOLTAGE ADJUST and CURRENT ADJUST controls to zero (full CCW), turn off Magnet Power Supply input power, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Leads Extensions. Lift and reseat the Ramp Leads. Repeat Steps 5 through 8.
9. Set the power supply VOLTMETER SELECT SWITCH to MAIN POWER SUPPLY position (This will display the output of the power supply monitored at the output lugs).

Note

Steps 10 through 12, below, are performed to check total system voltage drop. Voltages measurements higher than 2.2V can result in too much heat being conducted into the magnet. If the voltage exceeds 2.2V during the test, follow the procedures in Step 8 for adjusting contact resistance.

10. Gradually increase the VOLTAGE ADJUST control to pass 735A through the Main Power leads, Lead Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Lead Extensions.
11. Adjust the CURRENT ADJUST and VOLTAGE ADJUST controls to minimum (full CCW) and continue with the ramping procedure after completion of Step 10.



1.0T MAGNET POLARITY RAMPED NORMAL
ILLUSTRATION 9–2

9–3 RAMPING

Procedure:

CAUTION

If a Quench occurs during ramping, immediately turn VOLTAGE control and CURRENT control of the Main Coil Power Supply to zero (fully CCW). A quench is a rapid discharge of the magnetic field which will result in the rapid generation and expulsion of helium gas, rupturing the Burst Disc in the Vent System.

WARNING!

MAKE SURE THAT THE AXIAL SHIM SWITCH HEATERS ARE ON DURING RAMPING TO PREVENT COIL DAMAGE AND MAGNET QUENCH. GE POWER SUPPLIES HAVE PROTECTIVE CIRCUITRY TO PREVENT RAMPING VOLTAGE WITH THE AXIAL HEATER SWITCH IN THE "OFF" POSITION.

Note

Ice will form around the Ramp Lead Hold Down Tool Flow Holes during ramping. Remove ice as needed to maintain helium gas flow.

1. Make sure valve V2 is closed.

2. Make sure VOLTAGE ADJUST and CURRENTS ADJUST controls are at zero (full CCW).

Note

The Axial Switch Heater current is supplied by the Main Service Ramp Supply. The Transverse 1 and Transverse 2 Switch Heaters currents are supplied by the Shim Supply. On Mobile magnets, do not use the Axial Heater on the Shim Supply as this will increase boil off.

3. Set HEATER 2 SHIM AXIAL Switch, on the Main Power Supply, to 1 (ON).

9–3 RAMPING (continued)

4. Set the Ramp Supply MAIN POWER Switch to ON.
5. Adjust CURRENT ADJUST controls for maximum (full clockwise).
6. For Mobile magnets, set the Shim Power Supply HEATER 2 TRANSVERSE 1 and HEATER 3 TRANSVERSE 2 switches to 1 (on).
7. Set HEATER 1 MAIN switch to 1 (on). Wait 3 minutes.
8. Set the power supply VOLTMETER SELECT SWITCH to MAIN COIL position.

Note

The main coil driving voltage will decay from the initial setting of 4.0 V. This is normal and no attempt to readjust the power supply should be made until directed below.

9. Turn power supply VOLTAGE ADJUST control until a reading of 4.0 volts is observed on the power supply digital VOLTMETER.

Note

Measured inductance should be approximately 5.0 – 8.0 henrys. If the calculated value is outside this range, discontinue ramping and measure Main Coil Resistance (See FUNCTIONAL CHECKS, Section 3). Contact the Region MAC Team Representative.

10. Estimate the system inductance by measuring current change over a 10 second ramping interval:

L(inductance) = 10 x Voltage/Current Change

Note

This method will give inaccurate values of inductance when the current is less than 200 Amps.

Note

The Teslameter will lock on when magnet current rises between 350 to 400 amps (using a Range 5 Probe). This usually occurs between 0.6 and 0.7 Tesla (25.545900 MHz – 29.803550 MHz). If the teslameter has not locked on, after reaching a magnet current of 500 amps, slowly reduce the Main Coil Drive Voltage to approximately 0.0v, than select the "MANUAL" position on magnetometer and adjust the "COARSE" vernier knob until the teslameter locks on to the field. Once the teslameter is locked on, select "AUTO' mode and wait until the Teslameter locks, then continue with Ramping.

WARNING!

THE 1.0T MAGNET CAN BE QUENCHED IF LARGE CURRENT OR VOLTAGE CHANGES OCCUR NEAR PARKING. WHEN ADJUSTING THE CURRENT IN STEP 12 BELOW, TURN THE CURRENT CONTROL IN A SLOW, CONTINUOUS MOTION. BE CAREFUL NOT TO JERK THE CURRENT CONTROL.

11. When the magnet current reaches 740 amps, slowly turn the VOLTAGE ADJUST control counterclockwise, at a rate of 0.2 amps/minute, until the desired parking field range of 1.0025 – 1.0028 Tesla (42.682690 – 42.695463 MHz) is obtained.

9–3 RAMPING (continued)

12. Check Teslameter and slowly adjust the VOLTAGE ADJUST controls, as required, to bring Magnetic Field between 1.0025 – 1.0028 Tesla (42.682690 – 42.695463 MHz). The total current will be approximately 740 amps. Allow final field to stabilize. The last two digits on the Teslameter should be the only digits changing.
13. Maintain field at final setting for 5 minutes before proceeding to Step 14.

Note

Observe voltage (read on Power Supply Voltmeter with toggle switch in MAIN COIL position). When field / current stabilizes the voltage across the magnet terminals will stabilize at 0.00.

14. Turn off Main Switch Heater. Wait a minimum of 15 minutes for the switch to fully cool and go "persistent".
15. Record current, frequency and lead extension voltage values at which the switch went "persistent" in DATA SHEETS, Table 6–1.

WARNING!

MAKE SURE THAT THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 6–1. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.

Note

Check that Teslameter does not decrease more than 0.3 gauss (Approx. 1.3 KHz) as the VOLTAGE ADJUST control knob is turned to Zero. Only the last three digits on the Teslameter should change. If the field decreases more than 0.3 gauss, as the VOLTAGE control knob is turned, the field is not stable and VOLTAGE ADJUST control adjustment must be discontinued until field is stable.

9–3 RAMPING (continued)

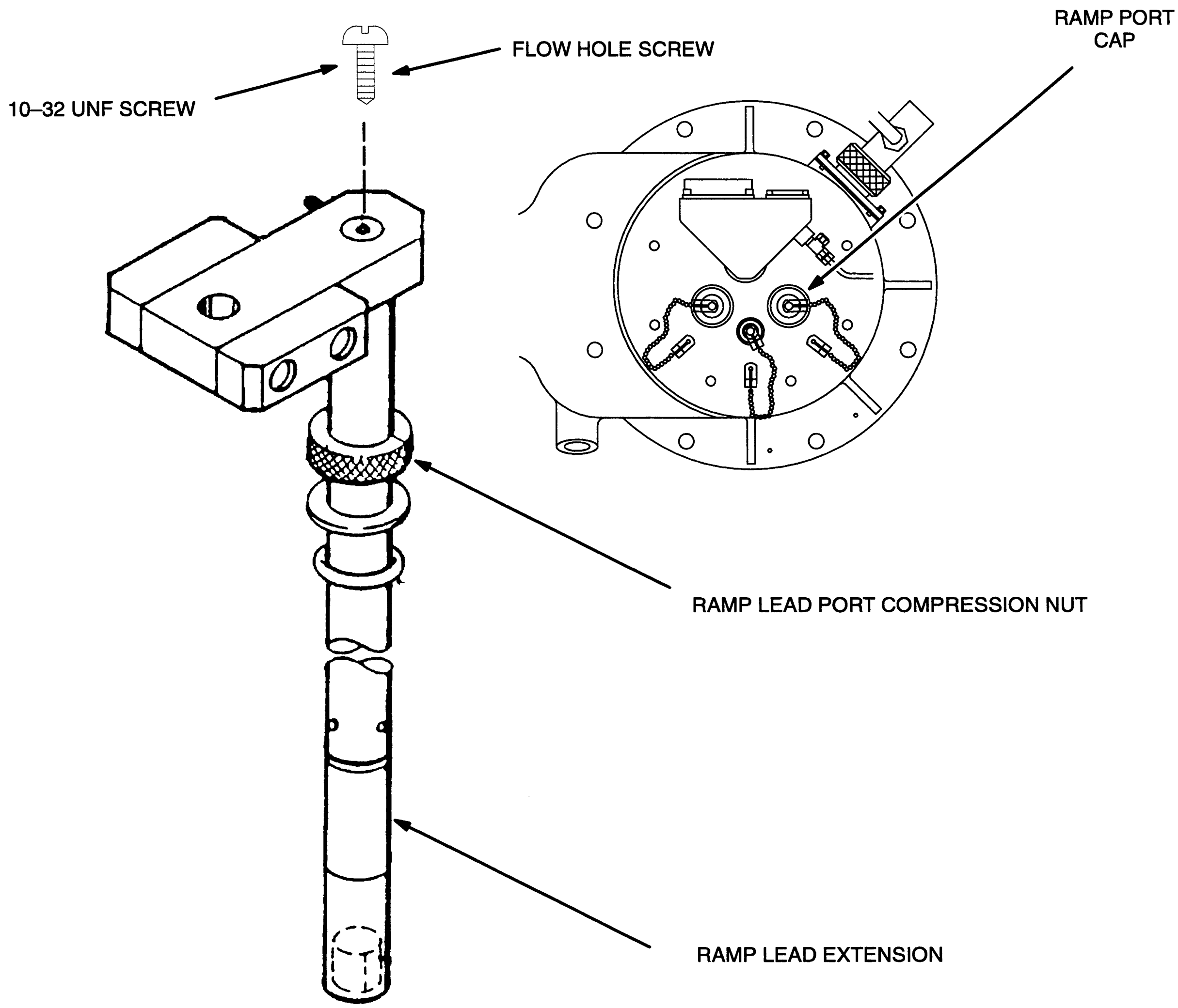
16. When the switch goes "persistent", slowly turn the power supply VOLTAGE ADJUST control to zero (full CCW) over a minumum of two minutes. The Main Coil Drive Voltage should not be greater than 3 millivolts as the VOLTAGE ADJUST control is adjusted. If this happens go back to Step 12 above because the magnet is not yet persistent.
17. Gradually turn the CURRENT ADJUST control to zero (over a one minute period).
18. Set HEATER 2 SHIM AXIAL switch to 0 (off).
19. For Mobile magnets, set the Shim Power Supply HEATER 2 TRANSVERSE 1 and HEATER 3 TRANSVERSE 2 switches to 0 (off).
20. Set MAIN POWER and POWER ON switches to OFF, on Main Power Supply and disconnect AC Input Power Cable.
21. For Mobile magnets, set the Shim Power Supply MAIN POWER switch to OFF.
22. Remove ramp lead hold down tool and replace bolts on lead assembly mounting plate.

CAUTION

Step 23 must be followed precisely in order to avoid an excessive heat load being applied to the magnet cartridge and a possible quench following a magnet ramp.

23. After parking the magnet and disconnecting main power supply, plug and remove one ramp extension at a time in the following sequence.
    a) Open valve (V2) to de–pressurize the cryostat to 0.25 psig. Close V2.
    b) Install a screw into the flow hole of only one of the ramp lead extensions. See Illustration 9–3.
    c) Remove all ice around the ramp lead port compression nut on the ramp lead extension that is being removed (i.e. the ramp lead extension that has the flow hole plugged in Step b). See Illustration 9–3.
    d) Unscrew the ramp lead port compression nut and remove the ramp lead extension from the magnet. Immediately replace the cap onto the ramp lead port.
    e) Repeat Steps b through e for the other ramp lead extension.

9–3 RAMPING (continued)



RAMP LEAD EXTENSION AND RAMP PORT COMPRESSION NUT
ILLUSTRATION 9–3

9–3 RAMPING (continued)

Note
Transverse coil currents will be removed in SET UP AND CALIBRATION, Section 10–2.

24. There may be small currents induced onto the Transverse Coils by the Ramping process. Make sure these currents are removed in conformance with SET UP AND CALIBRATION, Section 10–2 before Magnetic Probe Centering and Shimming the magnet.

CAUTION

Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve (V2).

Note
Read all flow rates from the bottom of the float (ball) on the flow meters.

Note
Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize (at least one day).

25. Make sure the following conditions are maintained. Re–check settings in three days and again after one week:

    INSTRUMENTATION LEAD FLOWMETER (F2) = 0.8 – 1.2 SCFH
    SHIM LEAD FLOWMETER (F1) = 1.8 – 2.2 SCFH
    CRYOSTAT GAUGE PRESSURE = 0.25 – 0.50 psig

Note
If flow rate through F2 is less than 0.8 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and "bubble test" all exhaust plumbing joints, relief valve and Shim Lead Connector. The vessel can be pressureized by either turning on the Correction Coil Heaters or turning off the Cold Head Compressor. Make sure V2 is fully closed. Repair any leaks.

26. Proceed to SET UP AND CALIBRATION, Section 10 ("Shimming Preparation/Field Stabilization").